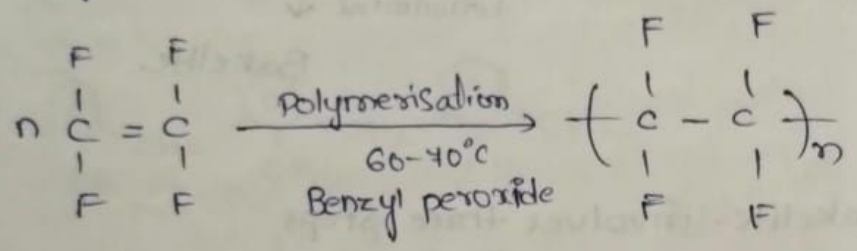
Uses :-

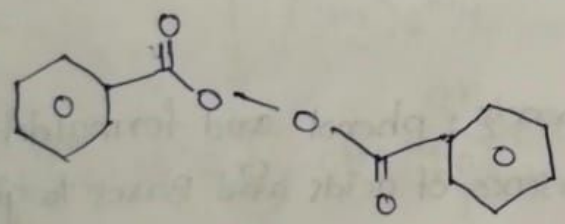
1. It is used for electrical insulation for covering of electrical cables. The production of pipes, sheets, and moulded parts
2. Tank linings, motor parts and safety elements
3. The production of toys handles. It is Also used for Jackets, Bags etc

Teflon

preparation:-

poly Tetra fluro ethylene [Teflon] is prepared by addition polymerisation of under pressure in the presence of Benzyl peroxide





→ Benzyl peroxide

properties

1. Teflon is extremely tough, flexible materials posessing high softening point [350°C]
2. It posesses extremely good electrical and mechanical properties
3. It is chemically resistant towards all chemicals
4. It has an ~~extreme~~ excellent thermal stability.

Uses :-

1. It is used for non stick plates
2. For making non stick surface coated utensils 3. In making Jackets, chemical carring pipes and tank linings
4. making bearings and burettes

5) $nCH_2=CH$ (OCOCH3) → $-[CH_2-CH(OCOCH_3)]_n-$

(Vinyl acetate) (poly vinyl acetate)

6) $nCH_2=CH(COOH)$ → $-[CH_2-CH(COOH)]_n-$

(Acrylic acid) (poly acrylic acid)

PVC (poly vinyl chloride) :-

preparation :-

preparation of PVC involves the following ~~three~~ two steps.

Step 1 : $HC\equiv HC + HCl \longrightarrow CH_2=CH(Cl)$

Vinyl chloride is prepared by treating of Acetylene with Hydrogen chloride at 60-80°C in the presence of metal chloride as catalyst.

Step 2 :-

PVC is prepared by Addition polymerisation of vinyl chloride in the presence of hydrogen peroxide catalyst and under pressure.

$$nCH_2=CH(Cl) \xrightarrow[H_2O_2,\ 60-70^\circ C]{\text{polymerisation}} -[CH_2-CH(Cl)]_n-$$

Vinyl chloride → Poly vinyl chloride.

properties :-

1. PVC is a colourless, odourless and chemically inert
2. It is insoluble in inorganic acids and Alkali
3. Specific Gravity – 1.35,
4. softening point 140°C

Plastics :-

plastics ~~are is~~ are a high molecular weight of organic molecule (or) materials, that can be moulded into any desired shape by the application of heat and pressure in the presence of catalyst

Advantages :-

1) They are light in weight. they possess low melting point.
2) They can be easily moulded and have excellent finished.
3) They posses good strength and toughness.
4) They are corossion resistance and chemically inert.

Disadvantages of plastics :-

1) They have high softness they undergo deformation under load
2) They undergo degradiation upon exposure to heat and U.V radiation.
3) combustibility is very high
4) They are non-biodegradable.

Classification of plastics.

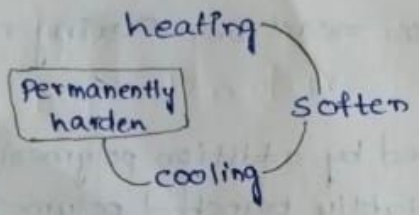
Based on the structures and type of resin used for manufacture of plastics, plastics are classified into two types

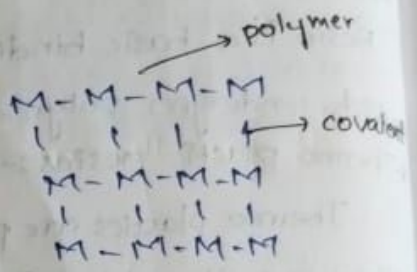
1. Thermo plastics
2. Thermo setting plastics.

Thermosetting resins:-

Thermosetting plastics are prepared by condensation polymerisation. Various polymer chains are held together by strong covalent bonds [cross-links]. Thermosetting plastics get hardened on ~~heating~~ cooling. Once harden they cannot be soften again.

Ex:- Nylon (6,6), Bakelite, PET





Differences between thermo plastic and thermo setting plastic

Thermo plastics	Thermosetting plastics.
1. plastics which are soften on heating and hardened on cooling are called Thermo plastics.	1. plastics which are hardened on ~~heating~~ cooling and cannot be soften by any means are called Thermo setting plastics
2. plasticity is reversible	2. plasticity is irreversible
3. These are formed by addition polymerisation.	3. These are formed by condensation polymerisation.
4. These are the linear in shape	4. These are cross-linked in shape
5. polymers have low molecular weight	5. polymers have high molecular weight
6. Weak soft and less brittle.	6. Strong hard and more brittle
7. These are soluble in organic solvents	7. These are insoluble in organic solvents.
8. Adjacent chains are held by weak wander wall forces	8. Adjacent chains are held together by strong covalent bonds

plastics

Thermoplastic	Thermosetting
PVC	Nylon(6,6)
PAN	Nylon 6
polyethylene	Bakellite
Teflon	PET

Resin :-

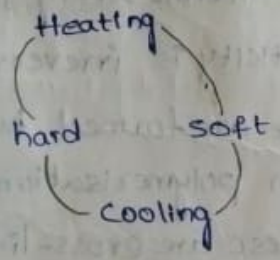
Resin is a basic binding material, present in plastics which undergoes polymerisation reaction. During moulding

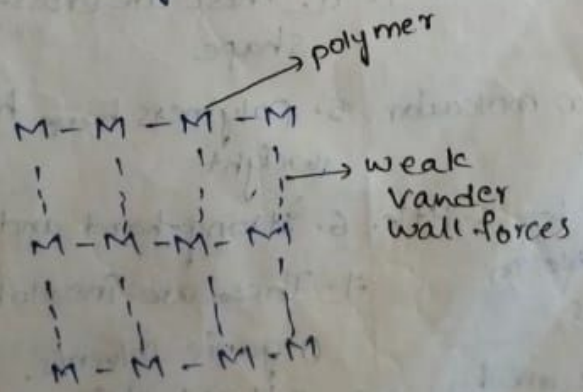
1. Thermo plastic resins :-

Thermo plastics are prepared by addition polymerisation They are straight chain or slightly branched polymers and various chains held are together weak wander wall forces of attraction.

Thermoplastics can be softened on heating and hardened on cooling

Ex:- PVC, PAN, TEFLON





9. Recycling is possible.	9. Recycling is not possible
10. Examples are PVC, PAN, Teflon	10. Examples for Nylon(6,6), Bakelite, PET

Differences between addition polymerisation and Condensation polymerisation

Addition polymerisation	Condensation polymerisation
1. In this polymerisation polymers are formed without elimination	1. In this polymerisation polymers are formed with elimination.
2. It is also called chain growth polymerisation	2. It is also called step growth polymerisation.
3. Monomers containing unsaturation undergoes in this polymerisation	3. Monomers containing functional groups undergoes this polymerisation.
4. Molecular weight of polymer is equal to some of monomers	4. Molecular weight of polymer is less than sum of molecular weight of monomers.
5. These are in linear shape	5. These are cross linked shape
6. Homochain polymer is formed	6. Heterochain polymer is formed.
8. Ex:- polyethylene, PVC, PAN, Teflon	8. Ex:- Nylon(6,6), Bakelite, PET
7. Thermoplastics are produced	7. thermo setting plastics are produced.

Rubbers (or) Elastomers :—

Rubbers (or) Elastomers are high polymers. [linear polymers], having elastic character

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Rubbers (or) Elastomers are high polymers. [linear polymers], having elastic character

Uses (or) applications :-

1. Oil & Gas industry
2. Aerospace
3. Food industry
4. Medical devices
5. Mining & constructions
6. consuming goods

Buna-S ...

Buna-S is a co-polymer formed by the co-polymerisation of styrene and 1,3 Butadiene.

eq in front page.

Properties :-

1. Good Flexibility : It has good flexibility over a wide ranges of temperature.
2. High tensile strength : SBR [Styrene Butadiene Rubber] has a decent strength making it useful in application that requires strength
3. Moderate heat resistance : It has moderate heat resistance, and its performs high temperatures.
4. poor oil resistance : SBR is not suitable for applications that requires resistances to oils and hydrocarbons.

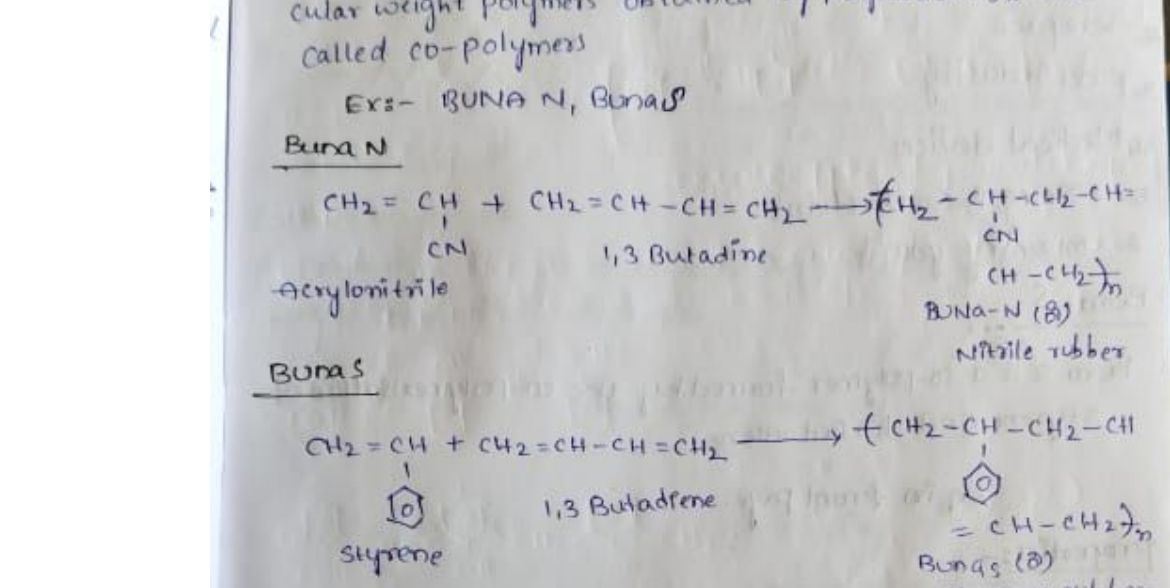
Uses (or) applications :-

1. Tires : SBR is used to key components in manufacturing of Tires, where it is used to thread compound.
2. Footwear : Buna-S is used to make a wide range of footwear including rubber boots, shoe soles to its flexibility and durability.

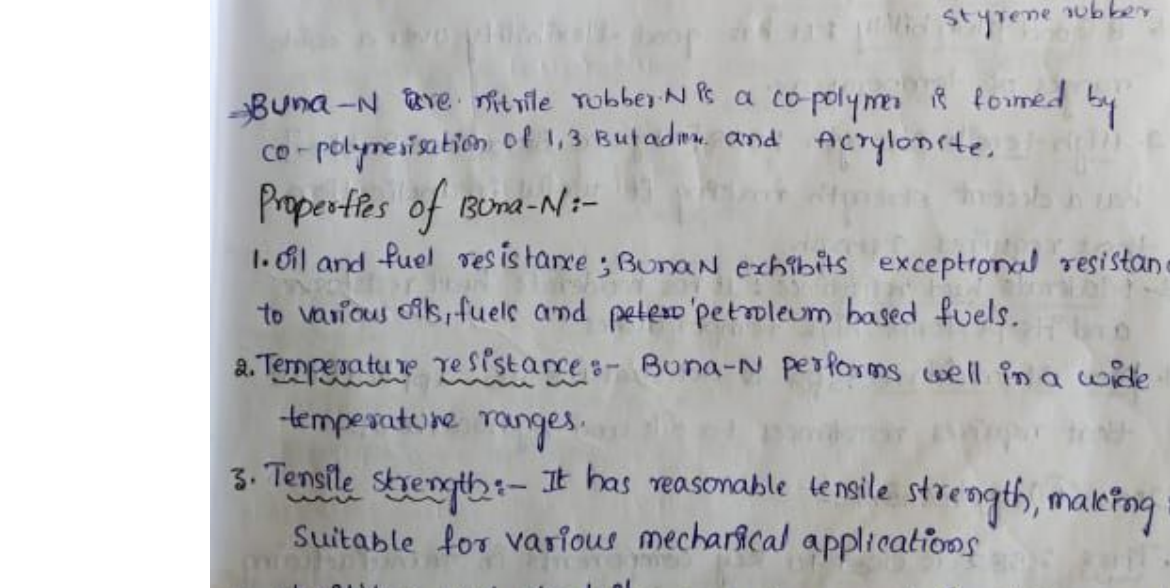
Co-polymerisation :- It is a joint polymerisation between two or different monomers combine to give polymer. High molecular weight polymers obtained by polymerisation are called co-polymers

Ex:- BUNA N, Bunas

Buna N



Bunas



→ Buna-N ~~are~~ nitrile rubber N is a co-polymer is formed by co-polymerisation of 1,3 Butadiene and Acrylonite.

Properties of Buna-N :-

1. Oil and fuel resistance ; BunaN exhibits exceptional resistance to various oils, fuels and ~~petro~~ petroleum based fuels.
2. Temperature resistance :- Buna-N performs well in a wide temperature ranges.
3. Tensile strength :- It has reasonable tensile strength, making it suitable for various mechanical applications
4. Flexibility and elasticity :- It has good flexibility and good elasticity, allowing to maintain ceiling property.
5. Electrical insulators :- It provides electrical insulat properties, making it suitable for some electrical and electronic applications.